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Propensity score matching–difference-in-differences analysis of the casual effect of opening intermediate high-speed railway stations on employment status in surrounding municipalities

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ABSTRACT

High-speed rail (HSR) has supplanted traditional rail as the dominant mode of intercity transportation due to its superior speed, punctuality, and convenience. However, the specific impacts of HSR development on municipalities surrounding intermediate stations remain insufficiently understood, particularly with respect to employment structures and local economic outcomes. This study employs propensity score matching to account for diverse characteristics of target municipalities, and combines this with a difference-in-differences framework to quantify the effects of HSR development. The results reveal clear regional heterogeneity: in less urbanized areas with higher shares of primary and secondary industries, HSR development is associated with declines in the secondary sector and growth in the tertiary sector. In contrast, in more urbanized regions, competition or substitution effects in the tertiary sector may limit such gains. Overall, while HSR construction can facilitate structural shifts toward tertiary sector-based economies, it does not consistently increase employment rates in municipalities surrounding new stations and may even contribute to higher unemployment in some regions, underscoring the need for regionally tailored development policies.

1. Introduction

The Shinkansen, the high-speed rail system in Japan, has been continuously developed since it first commenced operation in 1964. As of 2024, the system comprises 10 lines connecting 114 stations through a network of more than 3,000 km of railways. Future development plans include the extension of the Hokuriku Shinkansen to Osaka and the Hokkaido Shinkansen to Sapporo as shown in Fig. 1. Meanwhile, the Maglev Chuo Shinkansen is expected to link the Tokyo and Osaka metropolitan areas in approximately 1 h, forming a mega-city region with a population of 70 million people [1], which would become the world's largest metropolitan area. This mega-city region, interconnected by high-speed rail, has the potential to reshape Japan's demographic and economic development patterns, significantly promoting population and economic mono-polarization.

The development of high-speed rail infrastructure represents a critical nexus between transportation systems and socioeconomic dynamics in contemporary Japanese urban planning. As a pivotal component of national infrastructure, high-speed rail networks not only significantly enhance inter-urban transportation efficiency but also generate substantial systemic effects on regional economic vitality and labor market structures. The intricate interplay between infrastructure investment and socioeconomic transformation necessitates a nuanced scholarly examination of the potential correlations between transportation infrastructure development and employment patterns at the national level.

Regarding national employment trends in Japan, despite overall employment numbers showing consistent growth since 2012, the rapid aging of the population has resulted in a decline in the working-age population of individuals aged 15–64 years, while employment among those aged 65 years or older has increased substantially [2]. Moreover, compared with developed metropolitan areas such as Tokyo and Osaka, less-developed regional municipalities are experiencing labor shortages due to the outflow of the working-age population, and this is also a major problem faced by East Asian and similar countries around the world. The completion of high-speed rail infrastructure and the economic and transportation advantages it is expected to bring to

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[1] From *White Paper on Land, Infrastructure, Transport and Tourism in Japan* by Ministry of Land, Infrastructure, Transport and Tourism in 2021.
[2] From National Census.

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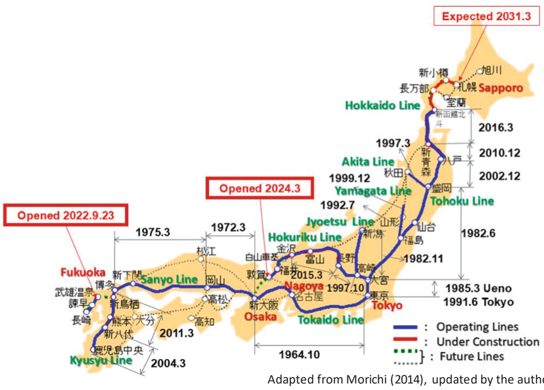

Adapted from Morichi (2014), updated by the authors.

Fig. 1. An Overview of the Development of High-Speed Rail in Japan.

metropolitan areas may attract working-age people from other municipalities along high-speed railway lines, potentially exacerbating the disparity between Japan's megacities and less-developed municipalities.

In an era where low-carbon transportation is being advocated worldwide, high-speed rail has garnered significant attention because of its safety, efficiency, and environmental sustainability. Although high-speed rail is expected to bring many economic benefits, if its development leads to negative externalities that affect municipalities along the routes, it may become necessary to implement targeted measures to mitigate these adverse impacts. Therefore, it is essential to analyze the specific effects of high-speed rail development on the employment conditions of local municipalities along these routes.

To explore the relationship between the construction of high-speed rail and changes in the employment situation in underdeveloped municipalities along the railway, this study employs causal inference methods (Rosenbaum and Rubin, 1983). To explore casual inference, the assumption of random assignment is fundamental and therefore necessitates randomized controlled trials. However, given that strict random assignment is impossible in observational studies and that confounding factors influencing both the presence of treatment and other covariates exist, numerous covariates must be introduced as independent variables to satisfy the strong ignorability assumption. But, directly estimating causal effects using high-dimensional independent variables in a sample is extremely challenging. Moreover, the impact of Shinkansen station openings is not limited to the opening period; it also has long-term, sustained effects.

Therefore, the longitudinal impact of Shinkansen station openings on each municipality along the route must be analyzed.

This study performs propensity score matching–difference-in-differences (PSM–DID) analysis (Rubin, 1983) with multi-period observational data to examine changes in socio-economic variables of interest in order to estimate the casual effects of high-speed railway station construction.

The remainder of this article is organized as follows: In Section 2, we review the literature to highlight the position and significance of this study. Section 3 introduces the analytical methods and variables. Section 4 presents and discusses the results of the causal inference. Finally, Section 5 presents the conclusions and future research directions.

2. Literature review and Positioning of this research

Since the establishment of the world's first commercially operated high-speed railway in Japan in 1964, high-speed rail networks have expanded worldwide. Japan and Europe pioneered relatively advanced high-speed rail systems, with China, South Korea, and Southeast Asian countries following closely behind, actively exploring and expanding high-speed rail systems (Moshe, 2006). In Japan's case, the construction of dedicated high-speed rail lines became the mainstream development approach; while operating costs are higher, these systems exhibit strong independence and avoid interference from other railway systems. In contrast, European models such as France's TGV, Germany's ICE, and other high-speed rail systems based on these two models chose a different developmental path from Japanese high-speed rail—upgrading existing railway lines. This approach offers greater flexibility and lower costs compared to dedicated railway lines, but sharing tracks with conventional trains introduces a series of operational and safety challenges (Anthony and Andrew, 2015). Additionally, regarding the development of Japanese high-speed railways, Okada (1994) provided an overview of the benefits and effects of high-speed rail, discussing the benefits of opening a high-speed rail station in terms of travel time, the impact on surrounding urban areas after opening, and environmental protection.

In recent years, an increasing number of studies have focused on the impact of transportation interventions on employment structure and residential location choices in surrounding areas. Pagliara and Wilson (2010) systematically reviewed the development of residential location choice models, emphasizing that such models are not only used to explain residential choice behavior but also constitute an important component of integrated land use and transportation models, providing a theoretical framework for analyzing how transportation infrastructure affects residential location and employment distribution by changing accessibility.

Many studies on high-speed railways have analyzed the effect of stations. For example, Chen et al. (2020) used principal component analysis and cluster analysis to clarify the impacts within a circular area centered around Shinkansen stations, and Terabe et al. (2016) used cluster analysis to explore the social and economic changes in the areas surrounding stations after the completion of the Kyushu Shinkansen. Meanwhile, Fuyama et al., (2017) compared changes in a region before and after the construction of a Shinkansen station.

Research on the impact of the development of conventional railway line stations on the population of municipalities has also been conducted (Sedef et al., 2011) and this literature empirically confirms the commonsense notion that transportation improvement enhances the development of a region.

Talebian et al. (2018) used propensity score matching (PSM) to evaluated the impact of California's financial support for Amtrak Stations on the local population and employment using PSM. Specifically, they conducted performed a multiple regression analysis by adjusting the balance of covariates, using PSM between the groups that received financial support and those that did not at the county and city levels. The results revealed that cities and counties where Amtrak stations were developed with financial support from the State of California became attractive residential areas for people and positively affected population growth. However, the impact on local employment remained limited. Jia et al. (2017) performed a comparative analysis of economic benefits and impacts across different high-speed rail lines. Through the application of difference-in-differences (DID) analysis and PSM–DID analysis, they demonstrated that high-speed rail development exhibits generally positive economic trends to a certain extent. However, the authors also emphasized that the economic impacts may vary substantially across different high-speed rail lines and may even produce the opposite effect.

In addition, some studies have suggested that the existence of high-speed railways does not always have a positive effect on municipalities along the route. Pagliara et al. (2016) and Biggiero et al. (2017)

examined the impact of high-speed rail (HSR) investment and construction on spatial equity within the Italian domestic high-speed rail system. Employing RP/SP (Revealed Preference/Stated Preference) surveys targeting Italian high-speed rail passengers, their research demonstrated that access and egress costs (time and expenses required to reach high-speed rail stations) significantly influenced passengers' primary mode choices. This effect was particularly pronounced for residents in low-income and relatively remote areas, resulting in geographical exclusion. Extending the analysis to the broader European context, Pagliara et al. (2022) used the Gini coefficient to analyze the benefits of high-speed rail in the UK, France, Spain, and Italy, finding that the development of high-speed rail in these countries had the potential to create social disparities. Therefore, it is important to ensure that the gap between urban and rural areas does not widen. Additionally, Cavallaro et al. (2022) considered construction costs and average travel times in analyzing the choices of railway users, showing that, for short-distance travel, the impact of high-speed rail was smaller than that of commuter rail. Thus, high-speed rail plays a more significant role in medium- and long-distance travel.

Previous studies have also analyzed the spatial extent of the development effects of other transportation infrastructures through post hoc evaluations. For example, Kunimi and Seya, (2021) developed a method within generalized synthetic control to define the extent of an area affected by the development of transportation infrastructure. Other studies have examined methods for calculating the impact range of transportation infrastructure development based on land price fluctuations and have utilized the development status of conventional railways to verify these methods (Seya et al., 2016). Another study used survival analysis to investigate the relationship between the size of the circular impact range centered on highway interchanges and the locations of factories in municipalities around the interchange (Kadokawa, 2019).

On the other side, Holl (2016) notes that medium to large cities with highways and interchanges are not randomly selected but rather exhibit endogeneity. Metropolitan areas and capital regions with higher transportation demands are prioritized for highway construction compared with small municipalities in rural areas. Consequently, highways are more likely to be constructed in regions seeking to enhance productivity and promote local economic development. In other words, this non-random selection of treatment group samples deviates from randomized experimental standards, potentially introducing selection bias and possibly leading to errors in causal inference.

Although existing research has explored the impacts of high-speed rail construction from multiple perspectives including economic development and demographic changes, empirical analysis at the more granular municipal level remains relatively insufficient in countries like Japan with diverse industrial structures and increasingly severe population aging, compared to prefecture-level analyses. In particular, systematic identification studies focusing on industrial structure and long-term employment changes are still relatively scarce. Therefore, this study, based on Rubin's potential outcomes framework for causal inference (Rubin, 1974), employs propensity score matching (PSM) combined with difference-in-differences (DID) methodology to conduct a systematic causal inference analysis of the impact of Shinkansen station construction on industrial structure and employment conditions in surrounding municipalities from 2000 to 2020.

Furthermore, to control for the interference of differences in industrial foundations across regions on policy effect estimation, this study identifies differences caused by regional heterogeneity through methods such as introducing interaction terms between industry types and years in the DID model. To more clearly position this study within the relevant empirical literature, Table 1 provides a systematic comparison of analytical units, methodological choices, and research variables across selected representative studies.

As shown above, existing empirical research on transportation infrastructure has widely employed methods including linear regression (OLS), geographically weighted regression (GWR), and various nonlinear models. While these methods can effectively describe spatial or nonlinear relationships between economic variables, they still have certain limitations in identifying the causal effects of transportation facility construction on surrounding areas. The propensity score matching combined with difference-in-differences (PSM-DID) methodology adopted in this study introduces time-individual fixed effects and industrial structure interaction terms based on ensuring policy intervention comparability, which is more conducive to revealing the heterogeneity of causal effects across different regions. The relevant content is summarized in Table 2.

Additionally, this study examines data from 2000 to 2020, comparing groups based on different high-speed rail station opening periods, thus achieving the first multi-period longitudinal analysis of station construction impacts on municipalities outside metropolitan areas in Japan. Regarding the definition of treatment areas, due to Japan's complex terrestrial topography with a high proportion of mountainous terrain, this study does not adopt the conventional fixed-radius impact zones centered on stations or definitions based on commuting

Table 1
Comparison of Selected Previous Studies and the Present Study.

Authors (Year)	Study Context	Method	Research Focus	Difference from This Study
Okada (1994)	Japan, Shinkansen	Descriptive study	Socioeconomic impacts of HSR	Does not involve systematic empirical analysis
Terabe et al. (2016)	Japan, Kyusyu Shinkansen	Cluster Analysis	Investigated social and economic changes near new or refitted Shinkansen stations.	Did not address causal effects, and the research scope was limited to the Kyushu region
Seya et al. (2016)	Japan, Transportation	Functional Data Analysis	Identification of the geographic scope of areas benefiting from transportation infrastructure	Focused primarily on spatial boundary identification rather than causal inference of transportation infrastructure construction effects
Jia et al. (2017)	China, High-speed Railway	PSM-DID	Causal impact of HSR on economic growth	Focuses on GDP growth, no sectoral employment analysis
Talebian et al. (2018)	USA, California, Amtrak	PSM + OLS Regression	Long-term impacts of intercity rail service on local population and civilian employment	Focus on population and overall employment, without sectoral employment structure or industry-specific effects analysis
Kunimi and Seya, (2021)	Japan, Tsukuba Express	Synthetic Control Method	Verification of the effects of transportation infrastructure construction on surrounding land prices	Due to different methodology (SCM), this study focuses on individual treatment effects rather than population-level effects

Table 2
Comparison of Empirical Methods for Causal Evaluation.

Method	Advantages	Limitations	Use in This Study or Not
Linear Regression (OLS)	Simple, interpretable, easy to implement	Susceptible to selection bias and unobserved heterogeneity	Not used in this study because it cannot effectively control for potential endogeneity and regional heterogeneity
Geographically Weighted Regression (GWR)	Accounts for spatial heterogeneity; reveals local variation patterns	Mainly descriptive; cannot identify causal effects over time	Not used because the study aims to identify time-series policy shocks causally
Non-linear/ML Approaches	Can model complex non-linear relationships; excels at prediction	Difficult to interpret variable effects; high complexity and limited extrapolation	Not used because the study focuses on causal inference rather than prediction
Synthetic Control Method (SCM)	Provides rigorous counterfactual for single or few treated units	Not suitable for many treated units or for analyzing heterogeneity across groups; high modeling complexity	Not used because the study involves multiple municipalities and group heterogeneity analysis
PSM-DID Analysis	Mitigates selection bias, improves comparability between groups, could introduce fixed effects and interaction terms to control sample heterogeneity	Heavily relies on the parallel trends assumption; results may be affected by unobserved confounding factors	Used as the primary analytical method, improving the accuracy of causal effect estimation by combining fixed effects and interaction terms

time. Instead, we employ the distance from each station to the administrative centers of nearby municipalities as the definition of treatment areas, thereby avoiding geographical bias while enhancing spatial accuracy through calibration with Open Street Map data. The methodology adopted in this paper is particularly suitable for medium-sized, mountainous industrialized countries such as Japan, South Korea, Italy, or Spain.

3. Research method

3.1. Subjects

This study intentionally excluded the Tokyo metropolitan area (Tokyo and the three surrounding prefectures), ordinance-designated cities, island areas, and evacuation areas in Fukushima Prefecture during the sample selection phase, aiming to avoid endogeneity bias caused by the strong employment attraction effects of large cities. Therefore, this analysis included 1,125 municipalities across Japan. Furthermore, to address potential selection bias arising from cities designated as terminal and originating stations during the pre-planning phase of high-speed rail development, we elected to exclude these municipalities from our study as well; this exclusion was implemented to eliminate the potential selection bias that might result from samples assigned to the treatment group prior to the actual treatment implementation.

3.2. Year Setting and assignment to treatment groups

The analysis period was 2000 to 2020. We focused on Shinkansen mainlines that were completed between 2000 and 2015, and we selected Shinkansen stations for analysis, with a focus on the opening years of these stations (Table 3). Consequently, municipalities in the Shikoku and Hokkaido regions, where Shinkansen lines had not been developed before 2015, were excluded.

Table 3
The number of subjects and shinkansen stations.

Group	A	B
Shinkansen Stations	Tōhoku Shinkansen (Iwate-Numakunai Station ~ Hachinohe station) Kyusyu Shinkansen (Shin-Yatsushiro Station ~ Sendai Station)	Tōhoku Shinkansen (Shichinohe-Towada Station) Kyusyu Shinkansen (Shin-Tosu Station ~ Shin-Tamana Station) Hokuriku Shinkansen (Iiyama Station ~ Shin-Takaoka Station)
Year of Opening	2000 ~ 2005	2010 ~ 2015
Number of municipalities	50	81

Based on the statistics of the variables used (e.g., the National Census), we identified two aggregate reference years: 2000 (2000–2005) and 2010 (2010–2015). These were divided into Group A and Group B, as shown in Table 3. Major intermediate and terminal cities were excluded from the analysis. The analysis years were set at 5-year intervals from the aggregation reference years, extending up to 2020. For example, in the case of the aggregation reference year 2000, the analysis years were 2000–2005, 2000–2010, 2000–2015, and 2000–2020. The aim was to analyze how the effects of the opening of Shinkansen stations changed over time.

To analyze the impact of Shinkansen station openings on the employment status in the surrounding municipalities, the station's sphere of influence must be defined. However, no consensus has been reached on the definition of the sphere of influence of Shinkansen stations; thus, various definitions exist. To facilitate analysis, we need to ensure sufficient sample size in the treatment group for matching. Considering that the minimum straight-line distance between Shinkansen stations in non-metropolitan areas exceeds the 70 km threshold, we decided to use municipal government offices (the main administrative centers of Japanese municipalities) as municipal centroids and set the treatment area within a 30-kilometer radius from Shinkansen stations to municipal government offices along regular road networks (excluding highways). This design can largely avoid spatial overlap between treatment and control groups of adjacent stations, thereby reducing the impact of direct spatial spillover effects on estimation results. Therefore, the Shinkansen station dummy equaled 1 for municipalities located within the defined 30-km range of a Shinkansen station that opened within 5 years of the aggregation reference year and 0 for those outside this range (Fig. 2). Fig. 3 and Table 3 shows the number of municipalities classified as having Shinkansen stations in each reference year.

3.3. Overview of propensity score

When the value of the allocation variable (whether the municipality has a Shinkansen station) is z_i and the value of the covariate (affecting both the allocation variable and result) of the i^{th} subject is x_i, then the municipality is in treatment group. The assigned probability e_i, called the propensity score, can be expressed by Equation (1) (Rosenbaum and Rubin, 1983).

$$e_i = p(z_i = 1|x_i) \tag{1}$$

Owing to the large number of independent variables, we could not feasibly use all the variables to directly match municipalities with Shinkansen stations to those without for a precise comparison. Therefore, we employed propensity scores to condense multidimensional independent variables into a single dimension for matching, thereby

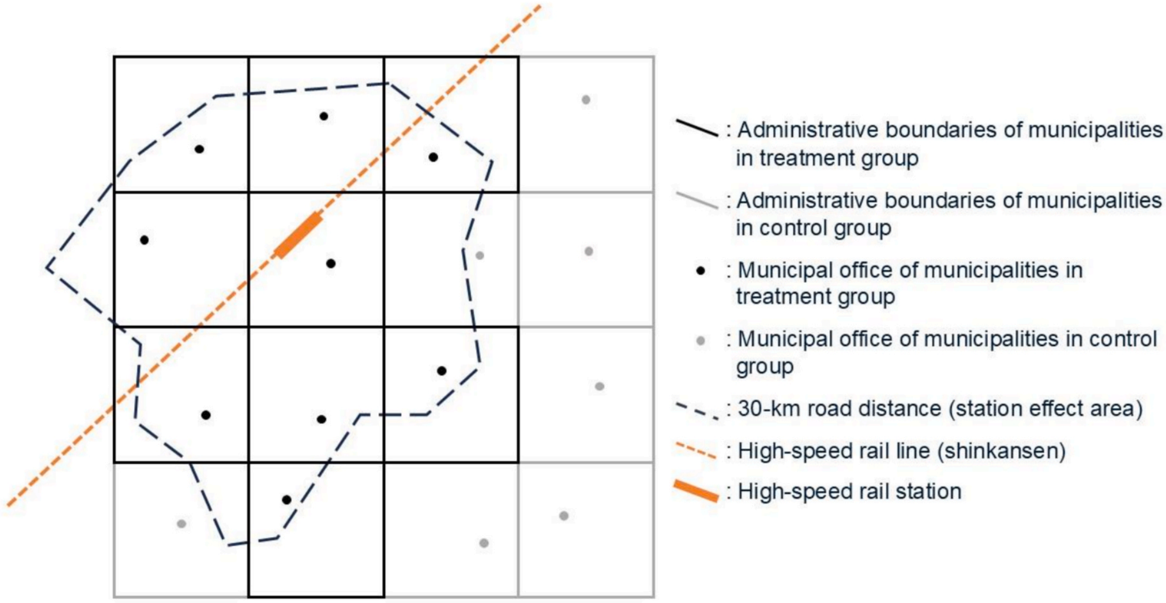

Fig. 2. The Sketch Map of Treatment Group.

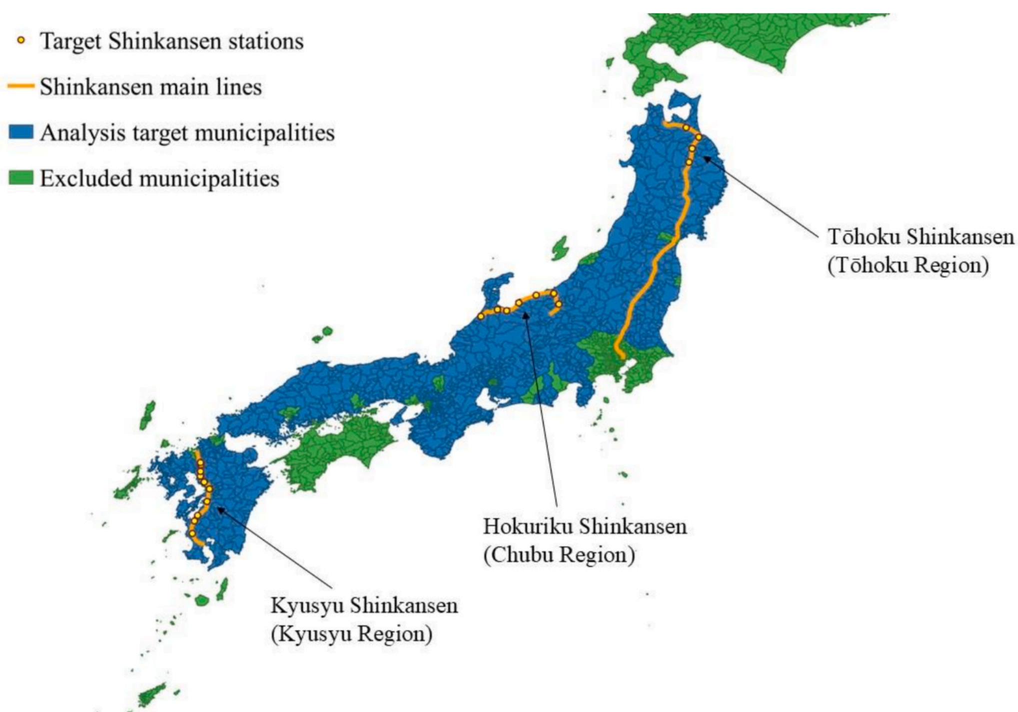

Fig. 3. Map of Target Subjects and Shinkansen Lines.

enabling the estimation of causal effects.

3.4. Variables used to calculate propensity score

The variables we used to calculate the propensity score are listed in Table 4. In the context of empirical research, variables are categorized into three distinct types: outcome variables, dependent variables, and independent variables. Outcome variables represent the quantitative metrics employed in DID analysis to evaluate the causal effects of high-speed railway infrastructure systematically. Because we used logistic regression to calculate the propensity score, dependent variables serve as binary indicators in the propensity score estimation to delineate the treatment and control groups while the independent variables function as covariates in the propensity score calculation, facilitating the comprehensive estimation of group-specific probabilities. All variables employed in this study are derived from original data sourced from official Japanese government statistics, representing the most detailed data available at the sample scale (municipality level) defined for this research.

To quantitatively analyze the causal effect of Shinkansen station openings on the employment status of municipalities around the station, we considered the following four outcome variables: 1) the proportion of employment in the secondary sector, which includes industrial production and manufacturing; 2) the proportion of employment in the

Table 4
Variables Used.

Variable name	Definition	Source
Outcome Variables		
Proportion of employment in the secondary industry	Number of secondary sector employees / total number of employees (from age 15 years)	e-Stat
Proportion of employment in the tertiary industry	Number of tertiary industry sector employees / total number of employees (from age 15 years)	e-Stat
Employment rate	Number of employed people/ Number of people aged 15 years and or order	e-Stat
Unemployment rate	Number of unemployed people / Number of people aged 15 years or older	e-Stat
Dependent Variable		
Shinkansen station *	Municipalities within a 30 km road distance of Shinkansen Station are counted as 1	Open Street Map
Independent Variables		
1.temp.H.Y	Annual maximum temperature	Meteorological mesh data
2.temp.L.Y	Annual minimum temperature	Meteorological mesh data
3.temp.A.Y	Annual average temperature	Meteorological mesh data
4.suns.t.Y	Annual average sunshine amount	Meteorological mesh data
5.ElevationAve.	Average altitude in the municipality	Altitude and slope mesh data
6.ElevationMax	Maximum altitude in the municipality	Altitude and slope mesh data
7.ElevationMin.	Minimum altitude in the municipality	Altitude and slope mesh data
8.SlopeMax	Maximum slope in the municipality	Altitude and slope mesh data
9.SlopeMin.	Minimum slope in the municipality	Altitude and slope mesh data
10.SlopeAve.	Average slope in the municipality	Altitude and slope mesh data
11.AnotherLA	Another land area/total area of the municipality	Land use mesh data
12.ForestLA	Forest area/total area of the municipality	Land use mesh data
13.BuildingLA	Building area/total area of the municipality	Land use mesh data
14.WastelandLA	Wasteland area/total area of the municipality	Land use mesh data
15.FarmingLA	Farming area/total area of the municipality	Land use mesh data
16.ConventionalLine	Length of the conventional line/ total area	Railroad time series data
17.Highway	Length of the highway/total area	Highway time series data
18.ConventionalLine Station	Number of the conventional line station in municipality area	Railroad time series data
19.Interchange	Number of the highway interchange in municipalities area	Highway time series data
20.NewTown	The number of New-towns in the municipality	New-Town data
21.Peopledummy *	The municipality with a total population of 30,000 or more as 1	National Census
22.Primary Industry Dominance *	The primary industry (agriculture etc.) accounts for the highest proportion as 1	National Census
23.Secondary Industry Dominance *	The secondary industry (construction etc.) accounts for the highest proportion as 1	National Census
24.Tertiary Industry Dominance *	The tertiary industry (commerce etc.) accounts for the highest proportion as 1	National Census

(*) dummy variable: Discrete variable with value 1 or 0

tertiary sector, which involves delivering services to businesses and consumers; 3) the employment rate; and 4) the unemployment rate. These outcome variables were sourced from e-Stat (The Portal Site of Official Statistics of Japan).

The dependent variable of the logistic regression in this study was the Shinkansen station dummy variable. To ensure the robustness of the propensity score model, this study comprehensively considers data spanning natural geographical attributes, socioeconomic indicators, and transportation infrastructure conditions. All independent variables are selected based on their potential influence on both high-speed rail station location decisions and industrial employment outcomes. Drawing upon the references discussed in **Chapter 2** and directly accessible official data published by the Japanese government, this study has determined to employ the following independent variables for propensity score regression calculations. Consequently, certain independent variables that present statistical challenges or are relatively abstract in nature, such as differences in local policies implemented across municipalities and cultural factors, are excluded from the analysis. The independent variables related to natural conditions and municipal land use include the variables 1 to 15. As Holl (2016) points out in his research, the location of transportation infrastructure is typically influenced by geographical conditions. Municipalities with proximate geographical conditions, such as those situated in similar latitudinal zones within plains or mountainous regions, tend to exhibit relatively similar probabilities of high-speed railway station construction, alongside comparably consistent patterns of economic development and social characteristics. In this context, independent variables related to temperature and sunshine hours are used to control for regions with similar climatic conditions. Independent variables associated with elevation and altitude are employed to control for samples located in mountainous regions versus those in plains, while variables related to slope reflect the degree of topographical variation in the region—a condition that directly affects station siting and indirectly influences industrial distribution. Neglecting the impact of slope could potentially introduce selection bias in geographical conditions between treatment and control groups. Furthermore, standardizing independent variables concerning slope or other aspects and calculating derived data for the entirety of Japan, such as weighted average slope, would be difficult to implement due to the overwhelming volume of data and issues of data incompleteness. Therefore, for independent variables related to geographical information, this research directly utilizes the maximum, minimum, and average data officially published by the Japanese government for calculations. Regarding the status of transportation infrastructure development, independent variables include variable 16 (the actual length of conventional railways divided by the total area of each municipality); variable 18, which is the number of conventional railway stations in each municipality; variable 17 (the actual length of highways divided by the total area of each municipality); and variable 19, which is the number of highway interchanges in each municipality. To account for the influence of nearby major cities on the analyzed municipalities, the number of "New Towns" is included as an independent variable (variable 20). New Towns is a Japanese term for strategically conceptualized urban settlements constructed in suburban peripheries as a comprehensive mitigation strategy to address metropolitan spatial congestion and demographic concentration. These planned urban development's arise through two primary implementation modalities: 1) institutionally orchestrated developments initiated by national agencies and local governmental entities, which are predicated on explicit legal frameworks such as the New Housing and Urban Development Projects and Land Readjustment Projects, and 2) autonomous developments carried out by private sector real estate developers operating under discretionary urban planning paradigms.

A municipal population dummy variable (variable 21) is used to distinguish between cities with large populations and towns or villages with smaller populations. This dummy variable is based on population data aggregated every 5 years from 2000 to 2020 (i.e., the national census). It covers municipalities with populations exceeding 30,000, as stipulated by the law concerning special provisions for municipal mergers, which categorizes such entities as cities. These dependent variables were sourced from e-Stat and the statistical site of Ministry of Land, Infrastructure, Transport and Tourism.

In addition, differences in industrial economic structures across municipalities may introduce estimation bias. Such differences often depend on whether municipalities have implemented specific industrial policies, as well as the content and timing of those policies, which are difficult to comprehensively and accurately quantify. Therefore, this study includes three industry-type dummy variables (variable 22, 23 and 24) to partially control for the influence of different industrial structures on employment and economic outcomes.

Using these variables, a logistic regression analysis was performed to obtain propensity scores, which represent the probability of each municipality being assigned to the treatment group. Following matching, we performed covariate balance tests and systematically optimized the covariate combinations to maximize the comparability between treatment and control groups, thereby minimizing potential confounding effects and enhancing the accuracy and reliability of causal inference. Additionally, this study employed the Hosmer-Lemeshow test and ROC curve analysis to evaluate the goodness-of-fit of the model, demonstrating that the combination of independent variables effectively predicts the allocation of treatment groups. Furthermore, through the integration of caliper matching and covariate balance diagnostics, we further ensured that the matched samples achieved balance across the selected independent variables. Further details are provided in subsequent sections.

3.5. *Strongly ignorable treatment assignment*

In randomized controlled trials, the causal effect of treatment can be directly ascertained through comparative analysis between the treatment and control groups, given the random allocation of samples that renders treatment assignment statistically independent of potential confounding factors. In contrast, in non-randomized experimental designs, treatment allocation is susceptible to underlying influential variables, compromising the complete randomization of individual assignments and potentially introducing confounding bias when attempting to observe causal effects through direct experimental results.

Thus, under the strongly ignorable treatment assignment, non-randomized experiments can, under certain conditions, be treated as randomized experiments, effectively eliminating confounding bias. By assuming that the treatment assignment y is random given covariates x, (i.e., that the potential outcomes y1 and y2 are independent of treatment assignment y conditional on x) and that for all possible values of the covariates, individuals have a non-zero probability of being assigned to either the treatment or control group, the treatment assignment y achieves conditional independence with respect to the potential outcomes. This relationship is formally expressed in Equation (2). (Rosenbaum and Rubin, 1983)

$$(y_1, y_2) \perp z | x, 0 < p(z = 1|x) < 1 \tag{2}$$

However, this condition cannot be directly verified in actual situations. But, if the assignment to the treatment and control groups can be explained by covariates, this condition can be indirectly verified. If the logistic regression model that predicts the presence or absence of

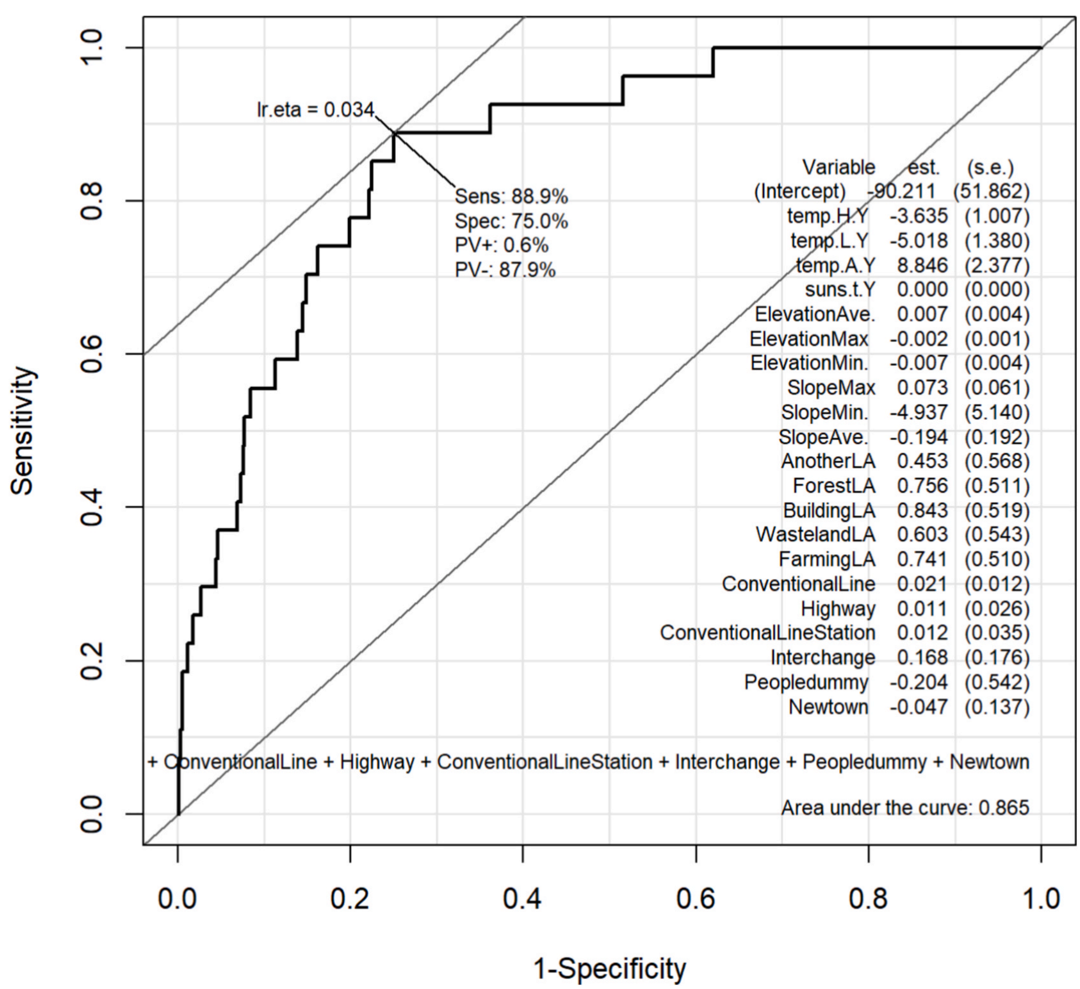

Fig. 4. The receiver operating characteristic curve of Group A.

treatment from the covariates fits well, the strongly ignorable treatment assignment condition can be considered satisfied. The model's goodness of fit was quantified using the C-statistic with a receiver operating characteristic (ROC) curve to verify whether this condition was satisfied.

The ROC curve is a tool for evaluating the goodness-of-fit of binary classification models and can be used to test whether propensity score models accurately predict the probability of individuals receiving treatment. If the area under the curve (AUC) is relatively large (above 0.8), it indicates that the classification model effectively captures key independent variables influencing treatment allocation. Conversely, if the AUC approaches 0.5, it suggests that the model cannot effectively differentiate between treatment and control groups, necessitating variable reselection. The vertical axis represents the true positive rate (Sensitivity, the proportion of treatment group samples accurately predicted as treatment group), while the horizontal axis represents the false positive rate (1-Specificity, the proportion of control group samples incorrectly predicted as treatment group). As shown in the Figs. 4 and 5 below, the closer the curve is to the upper left corner, the higher the true positive rate of the model's predictions, indicating better classification performance. Conversely, the closer the curve is to the 45° line, the higher the false positive rate, indicating poorer classification performance.

3.6. Difference-In-Differences analysis and its extension

DID analysis is used to compare outcomes before and after the implementation of a policy (Lindner and McConnell, 2019). This method considers the difference in outcomes between the pre- and post-implementation periods in the control group as an effect of time. The time effect is then subtracted from the difference in outcomes between the pre- and post-implementation periods in the treatment group in order to estimate the policy effect. To estimate the average treatment effect of the treatment group, the parallel trends assumption must be held. DID analysis is performed after using PSM to confirm that two municipalities in the same pair had the same development trend.

By calculating the objective variables (e.g., employment rate) in the treated and untreated municipalities, the average treatment effect on the treatment group, using the regression equation of the DID analysis, can be estimated using Equation (3).

$$y = \alpha_1 + \alpha_2 G + \alpha_3 T + \alpha_4 GT + \varepsilon \tag{3}$$

Here, y is the employment rate, G is the treatment dummy (whether the municipalities have a Shinkansen station or not), T is the time point dummy (a dummy variable that equals 0 for the reference year and 1 for 5 years after the reference year), ε is the error term, and α_1 to α_4 are coefficients. G and T are not related.

Each coefficient represents the following: α_1 denotes the average outcome variable of the control group before the treatment; α_2 captures the change in the average outcome variable of the control group before and after the treatment; α_3 represents the pre-treatment difference between the treatment and control groups; and α_4 indicates the causal effect of the treatment, specifically, the causal impact of the opening of Shinkansen stations on changes in the regional population, as estimated by the DID analysis. Based on Equation (3), we constructed a regression equation for DID analysis and calculates the average treatment effect on the treated (ATT).

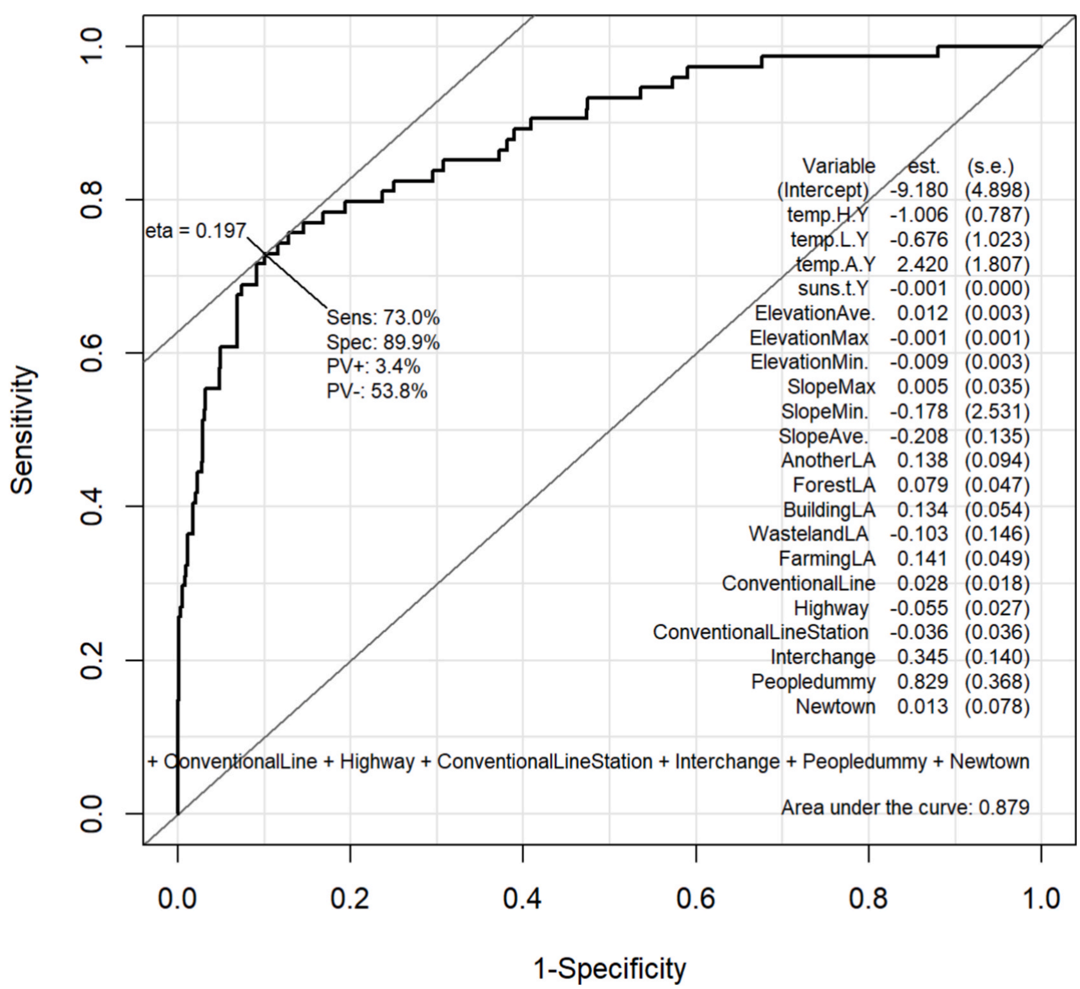

Fig. 5. The receiver operating characteristic curve of Group B.

The subjects of analysis in this study are Japanese municipalities, whose industrial structures significantly influence long-term trends in employment dynamics. For instance, municipalities dominated by secondary industries are susceptible to the effects of domestic industrial restructuring and global manufacturing cycles, while those dominated by tertiary industries may be influenced by major demographic trends such as population aging. Failure to account for the temporal variation induced by these industrial structural differences may violate the parallel trends assumption in the post-treatment period, thereby introducing bias in the estimation of Shinkansen station construction effects. Therefore, this study incorporates interaction terms between industry type dummy variables and year variables in the regression model to control for heterogeneous temporal trends across different industrial types (Kabore and Rivers, 2023). Furthermore, the inclusion of industry type-year interactions helps absorb systematic shocks that affect different industries uniformly (such as economic crises), thereby enhancing the credibility of the parallel trends assumption. This specification also partially mitigates potential estimation bias arising from the correlation between industrial structure and Shinkansen site selection, contributing to more robust identification of the causal impact of Shinkansen construction on employment.

Furthermore, although we excluded special samples such as metropolitan areas to prevent their unique economic and functional characteristics from exerting disproportionate influence on the analytical results, we also recognize potential heterogeneity issues related to regional attributes within the retained sample. To address temporal heterogeneity arising from different industrial structures, we introduced interaction terms between industry-type dummy variables and year variables. Additionally, we included municipality and year fixed effects to account for time-invariant regional characteristics and common macroeconomic shocks. This study can effectively identify the causal effects of Shinkansen construction on employment within the analytical sample, even with the exclusion of certain special regions. However, we acknowledge that caution should be exercised when extrapolating these findings to the excluded regions.

Therefore, Equation (4), which is developed by adding interaction terms to Equation (3), is as follows:

$$y = \alpha_1 + \alpha_2 G + \alpha_3 T + \alpha_4 GT + \beta_1(Pri.Ind. \times year) + \beta_2(Sec.Ind. \times year) + \beta_3(Ter.Ind. \times year) + \delta_i + \mu_i + \varepsilon \tag{4}$$

Here, *Pri.Ind.*, *Sec.Ind.*, and *Ter.Ind.* are dummy variables reflecting the dominant industries of each municipality as defined in the independent variables section of Chapter 3, Section 4, and *year* represents the year variable, δ_i denotes the year fixed effects, μ_i denotes the municipality fixed effects.

4. The average treatment effect on treated of Shinkansen stations construction

4.1. The covariate balance

To accurately verify the causal effects, we need to ensure that the independent variables of the treatment and control groups matched in the PSM have similar Euclidean distances in order to achieve covariate balance between the two groups (Hansen and Bowers, 2008). Although we can ensure covariate balance to a certain extent through matching, it is necessary to calculate the absolute value of the standardized mean difference. The covariate balance used in this study is shown in Figs. 6 and 7.

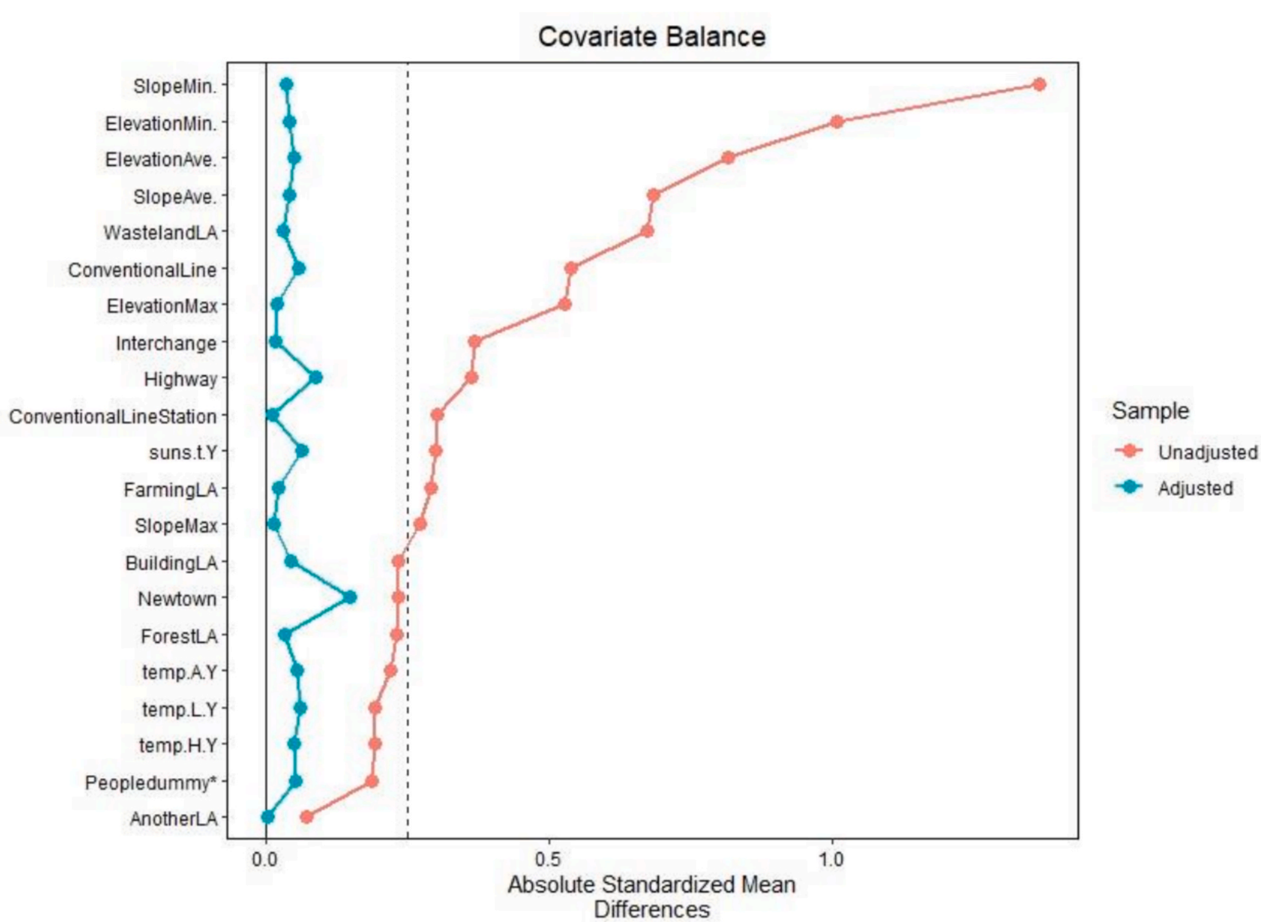

Fig. 6. The Covariate Balance of Group A.

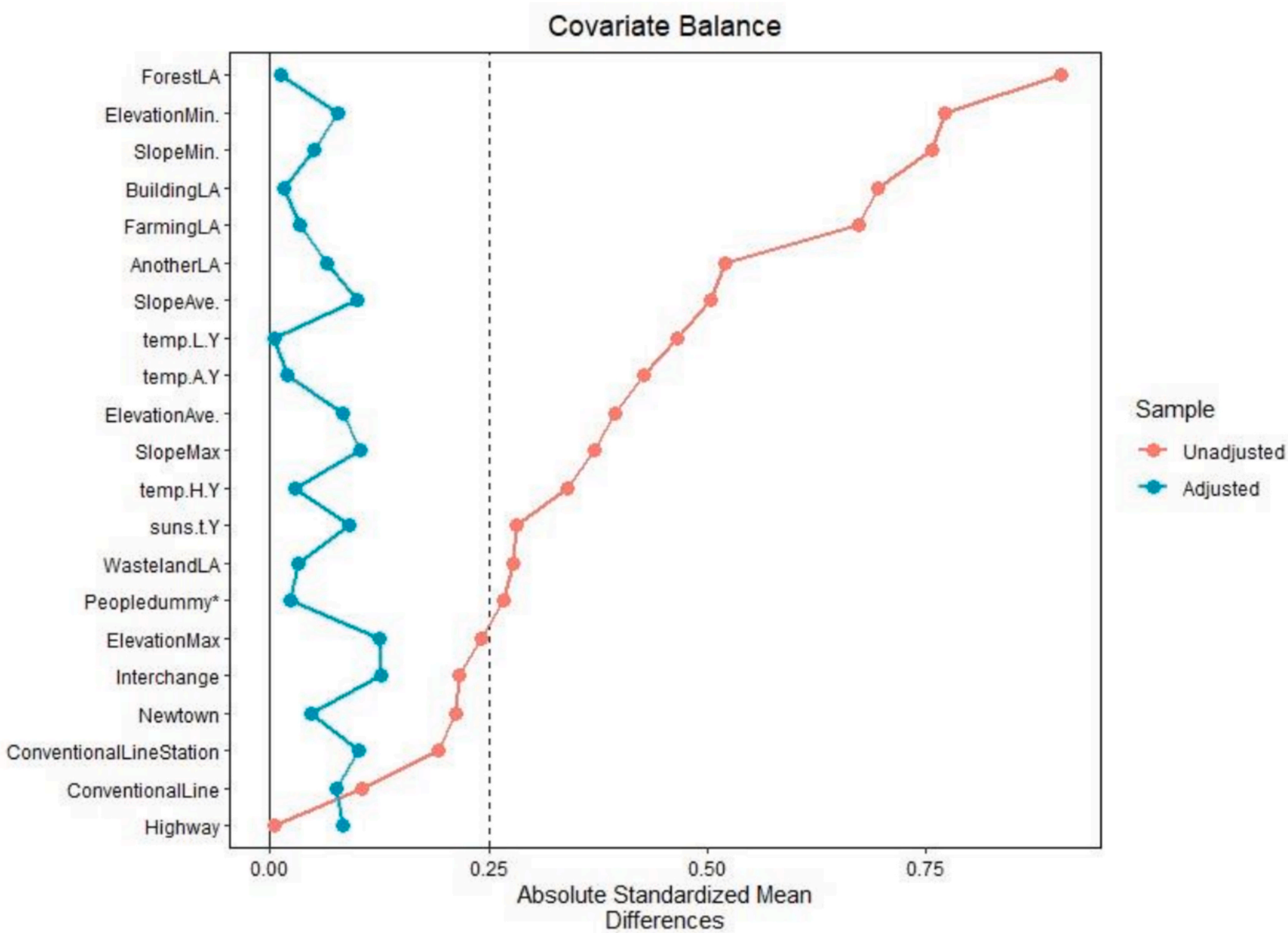

Fig. 7. The Covariate Balance of Group B.

The orange line represents the differences between the treatment and control groups for a given covariate before adjustment, while the blue line represents the differences after adjustment (i.e., after calculating the propensity scores). As shown in Figs. 6 and 7, there are substantial differences between the treatment and control groups in covariates such as minimum slope and land-use ratio prior to adjustment, indicating a failure to meet the assumptions of a randomized controlled trial. However, after adjustment, the differences between the treatment and control groups across all covariates are significantly reduced, achieving a balanced state.

4.2. Propensity scores

As mentioned above, we calculated propensity scores using logistic regression based on Table 4 and validated their significance. Table 5 presents the logistic regression model employed in this calculation, including the coefficients for each covariate, results of the Hosmer–Lemeshow (HL) test (Hosmer and Lemeshow, 2001) for the logistic regression model, and the C-statistic, which was introduced in the preceding section.

Because the goodness of fit for a logistic regression cannot be directly determined by the significance of the parameters for each covariate, we used the HL test. This method divides the population into multiple subsets, calculates the difference between the observed and expected values for each subset, and uses the differences from each group to calculate the HL statistic. Because the HL statistic follows a chi-square distribution, we can easily derive the p-value for the HL test. When

Table 5
The Result of the Propensity Score and Test.

Group	A		B	
Independent variables	Estimate	Pr(>|z|)	Estimate	Pr(>|z|)
(Intercept)	−90.2100		−9.1802	
temp.H.Y	−3.6350	***	−1.0055	
temp.L.Y	−5.0180	***	−0.6762	
temp.A.Y	8.8460	***	2.4199	
suns.t.Y	0.0003		−0.0007	***
ElevationAve.	0.0066		0.0117	***
ElevationMax	−0.0020		−0.0007	
ElevationMin.	−0.0072		−0.0087	**
SlopeMax	0.0730		0.0047	
SlopeMin.	−4.9370		−0.1782	
SlopeAve.	−0.1937		−0.2076	
AnotherLA	0.4527		0.1380	
ForestLA	0.7556		0.0791	
BuildingLA	0.8435		0.1344	*
WastelandLA	0.6029		−0.1034	
FarmingLA	0.7410		0.1411	**
ConventionalLine	0.0209		0.0282	
Highway	0.0105		−0.0547	*
ConventionalLineStation	0.0119		−0.0356	
Interchange	0.1681		0.3449	*
Peopledummy	−0.2045		0.8290	*
Newtown	−0.0472		0.0135	
Hosmer-Lemeshow Test	0.918		0.316	
c statistics	0.865		0.879	

Significance: *** 0.1%, ** 1%, * 5%.

the p-value is greater than 0.05, which is not significant, we cannot conclude that the logistic regression model has a poor goodness of fit. Therefore, when the p-value is greater than 0.05, we can conclude that the model has a good goodness of fit.

4.3. Matching

We used the non-replacement nearest-neighbor matching method (Caliendo and Kopeinig, 2008). The control group municipalities with propensity scores closest to those of the treatment group municipalities were selected to form one-to-one pairs. This means that if the propensity scores of the treatment and control group municipalities are the same, perfect matching can be achieved. However, given that none of the treatment and control groups had equal scores, some bias was unavoidable during matching.

To minimize bias to the extent possible, we used caliper matching, which was performed based on nearest-neighbor matching and with the caliper set to 0.2. The distribution of propensity scores for the treatment and control group municipalities after caliper matching is shown in Figs. 8 and 9, respectively. In Group A, 24 pairs formed in the matching, while in Group B, 56 pairs formed.

4.4. Overview of the matched treatment Group and control Group pairs

The average values of the outcome variables for the treatment and control group municipalities involved in propensity score matching is shown in Figs. 10 to 13 for the treatment group and control group of Group A and Group B. Overall, the outcome variables for the treatment and control groups within the same group exhibit similar trends. Regarding the proportion of employment in secondary industries, we observed a continuous decline in Group A from 2000 to 2015, followed by a slight increase from 2015 to 2020. In Group B, this outcome variable showed little change from 2010 to 2020, with the control group having a higher average value than the treatment group. The trend of the proportion of employment in the tertiary industry is opposite to that in the secondary industry, showing a continuous increase. Additionally, the average value for the treatment group in Group A was lower than that for the control group.

Next, for the outcome variables of employment rate and unemployment rate, all four employment rate curves exhibited a noticeable dip around 2010, while all four unemployment rate curves peaked around the same year. This may be due to the 2008 financial crisis, which reduced employment rates and increased unemployment rates. Furthermore, the employment rate of the control group is generally higher than that of the treatment group, except in 2010, the unemployment rate curves show that the treatment group's unemployment rate is higher than that of the control group, except in 2000 and 2005. Additionally, the average values of the outcome variables are not based on the results of the DID analysis. To determine the causal effects of Shinkansen station construction, which is the focus of the DID analysis, we must calculate the difference in slopes between the treatment and control group curves in the same group.

4.5. The casual effect of Shinkansen station construction

Because the groups are classified according to the Shinkansen construction years and the areas where Shinkansen stations were built varied by construction year, the analysis of the results is region-specific. As shown in Table 3, the analysis for Group A targets the following areas: 1) Iwate-Numakunai Station to Hachinohe Station on the Tōhoku Shinkansen, which serves the northeastern part of the Tōhoku region; and 2) Shin-Yatsushiro Station to Sendai Station on the Kyushu Shinkansen, which serves the southern part of the Kyushu region. Meanwhile, the analysis for Group B targets the following areas: 1) Shichinohe-Towada Station on the Tōhoku Shinkansen, serving the northeastern part of the Tōhoku region in Japan; 2) Shin-Tosu Station to Shin-Tamana Station on the Kyushu Shinkansen, serving the northern part of the Kyushu region; and 3) Iiyama Station to Shin-Takaoka Station on the Hokuriku Shinkansen, serving the northern part of the Chubu region. The ATT values derived from the DID analysis (i.e., the construction effects of the Shinkansen stations) are listed in Table 6.

4.5.1. Impact of Shinkansen station construction on secondary industry employment share

As shown in Table 6, the ATT estimates for Group A are consistently negative across all periods, indicating a statistically significant association between Shinkansen station construction and declining secondary industry employment share in northeastern Tōhoku and southern Kyushu. The estimation results for Group B show weakly positive values close to zero, suggesting minimal or no discernible impact of Shinkansen construction on secondary industry employment share in northern Kyushu and northern Chubu. Given that the industrial base of Group A regions is predominantly traditional manufacturing and has been significantly affected by global manufacturing competition and domestic structural adjustment in recent years, this difference can be partially explained as heterogeneity in Shinkansen construction effects across different industrial structures.

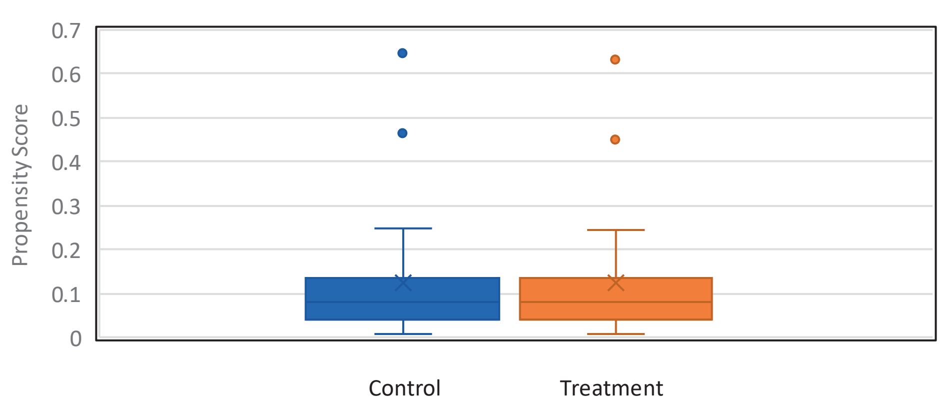

Fig. 8. Distribution of propensity scores in Group A.

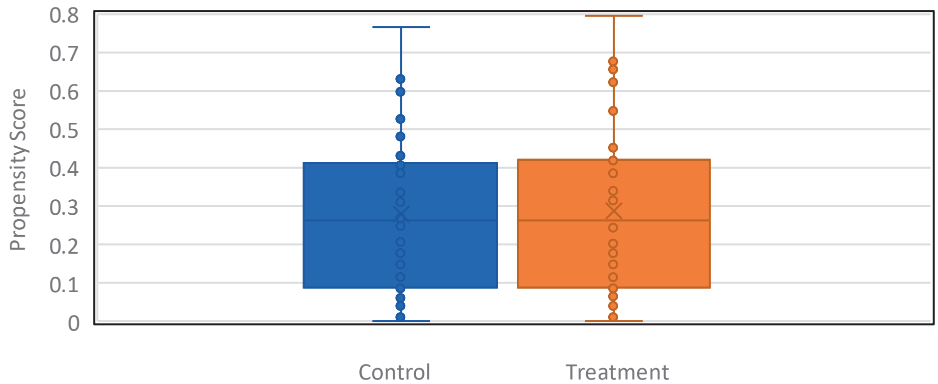

Fig. 9. Distribution of propensity scores in Group B.

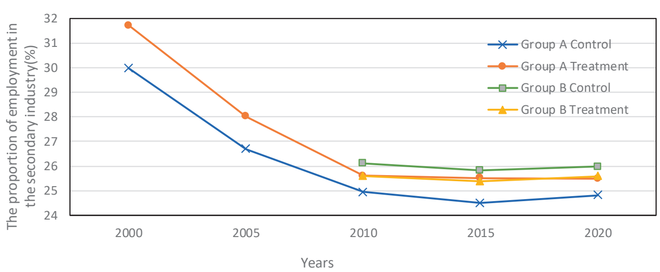

Fig. 10. Average of the proportion of employment in the secondary sector.

4.5.2. *Impact of Shinkansen station construction on tertiary industry employment share*

As shown in Table 6, the ATT estimates for Group A are consistently positive across all periods, suggesting that Shinkansen construction in northeastern Tōhoku and southern Kyushu may have promoted growth in tertiary industry employment share. In contrast, the ATT estimates for Group B are negative, possibly reflecting strong service industry competition or substitution effects in northern Kyushu and northern Chubu regions, or failure to achieve anticipated service industry growth post-opening. The contrast between the two groups highlights the heterogeneity in industrial structure and Shinkansen economic effects across different regions.

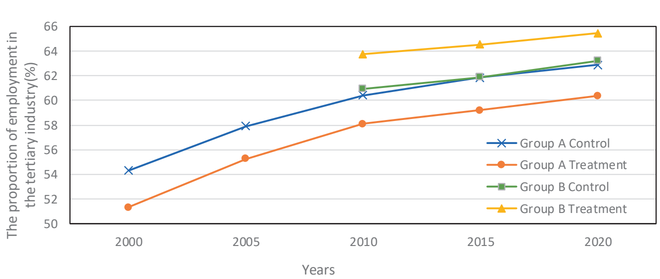

Fig. 11. Average of the proportion of employment in the tertiary sector.

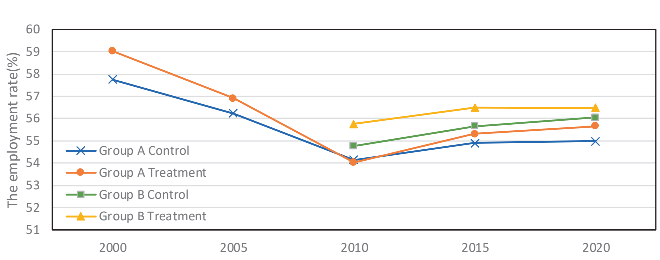

Fig. 12. Average employment rate.

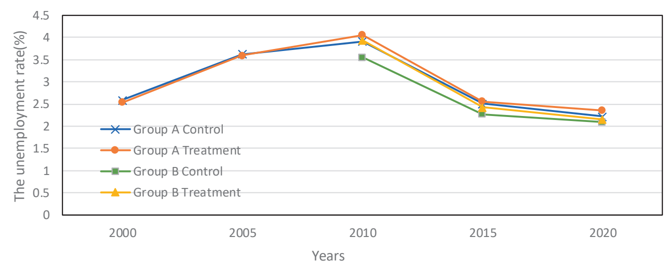

Fig. 13. Average unemployment rate.

4.5.3. *Impact of Shinkansen station construction on employment rate*

Table 6 shows that ATT estimates for both Group A and Group B are negative, suggesting that employment rate growth in Shinkansen station construction areas did not significantly exceed that of control areas, and was even estimated to be slightly lower than the control group. This result requires cautious interpretation. Possible explanations include: labor outflow due to improved transportation conditions, or employment contraction in certain sectors during economic structural transformation induced by Shinkansen construction.

4.5.4. *Impact of Shinkansen station construction on unemployment rate*

As shown in Table 6, estimation results for some periods in Group A are not statistically significant. We observe a trend of increasing unemployment rates, which may reflect economic structural adjustment pressures or increased labor market mobility accompanying Shinkansen opening. The ATT estimates for Group B are negative, indicating that Shinkansen construction in northern Kyushu and northern Chubu may have facilitated employment matching or generated employment opportunities to some extent, thereby reducing unemployment rates. The differences between group results suggest that Shinkansen effects may vary according to regional characteristics, industrial structure, and population dynamics.

5. Conclusion

This study employs a propensity score matching combined with difference-in-differences (PSM-DID) methodology to evaluate the causal effects of Shinkansen station construction on local employment dynamics across Japanese municipalities from 2000 to 2020. By controlling for geographic, socioeconomic, and industrial structural characteristics, and considering the heterogeneous effects between two groups of regions that differ in opening periods and regional economic contexts, the DID model incorporates time fixed effects, individual fixed effects, and interaction terms between industry types and years.

Moreover, the methodological framework employed in this study—combining propensity score matching, difference-in-differences

Table 6
ATT of Shinkansen Stations.

ATT of the proportion of employment in the secondary sector in Group A				
Years	2000–2005 (Within 5 years of opening)	2000–2010 (Within 10 years of opening)	2000–2015 (Within 15 years of opening)	2000–2020 (Within 20 years of opening)
ATT	−0.396	−1.067	−0.772	−1.077
Std.Error	0.936	2.264	2.248	2.337
t-value	−0.115	−0.471	−0.343	0.047
p-value	1.006e-04	2.378e-06	7.154e-07	1.315e-06
ATT of the proportion of employment in the secondary industry sector in Group B				
Years			2010–2015 (Within 5 years of opening)	2010–2020 (Within 10 years of opening)
ATT			0.016	0.027
Std.Error			1.715	1.875
t-value			0.009	0.064
p-value			5.505e-07	2.614e-06
ATT of the proportion of employment in the tertiary sector in Group A				
Years	2000–2005 (Within 5 years of opening)	2000–2010 (Within 10 years of opening)	2000–2015 (Within 15 years of opening)	2000–2020 (Within 20 years of opening)
ATT	0.344	0.653	0.319	0.392
Std.Error	3.097	3.095	3.017	2.989
t-value	0.111	0.211	0.106	0.188
p-value	2.134e-07	7.634e-09	3.022e-10	5.612e-11
ATT of the proportion of employment in the tertiary sector in Group B				
Years			2010–2015 (Within 5 years of opening)	2010–2020 (Within 10 years of opening)
ATT			−0.149	−0.499
Std.Error			2.474	2.455
t-value			−0.060	−0.203
p-value			1.383e-4	7.287e-5
ATT of the employment rate in Group A				
Years	2000–2005 (Within 5 years of opening)	2000–2010 (Within 10 years of opening)	2000–2015 (Within 15 years of opening)	2000–2020 (Within 20 years of opening)
ATT	−0.567	−1.302	−0.789	−0.514
Std.Error	1.301	1.269	1.262	1.321
t-value	−0.436	−1.026	−0.625	−0.389
p-value	4.682e-07	2.982e-11	2.549e-09	8.272e-08
ATT of the employment rate in Group B				
Years			2010–2015 (Within 5 years of opening)	2010–2020 (Within 10 years of opening)
ATT			−0.202	−0.628
Std.Error			1.080	1.072
t-value			−0.187	−0.586
p-value			2.088e-05	2.629e-05
ATT of the unemployment rate in Group A				
Years	2000–2005 (Within 5 years of opening)	2000–2010 (Within 10 years of opening)	2000–2015 (Within 15 years of opening)	2000–2020 (Within 20 years of opening)
ATT	0.0002	0.164	0.097	0.171
Std.Error	0.287	0.280	0.197	0.193
t-value	0.001	0.586	0.495	0.691
p-value	1.073e-08	3.801e-14	0.516	0.010
ATT of the unemployment rate in Group B				
Years			2010–2015 (Within 5 years of opening)	2010–2020 (Within 10 years of opening)
ATT			−0.231	−0.315
Std.Error			0.223	0.214
t-value			−1.041	−1.475
p-value			2.20E-16	2.20E-16

estimation, municipality and year fixed effects, and interaction terms between industry types and years—provides a flexibly applicable paradigm for research in other contexts. Researchers and policymakers can utilize this methodology to evaluate infrastructure or policy interventions in regions with diverse industrial structures and population dynamics, thereby more precisely identifying and interpreting heterogeneous impacts across different areas.

The findings reveal significant inter-regional heterogeneity in the impact of Shinkansen station construction on local labor markets. Group A, located in northeastern Tōhoku and southern Kyushu, is characterized by higher proportions of primary and secondary industries, lower urbanization rates, and persistent labor force outflows. The results show that ATT estimates for secondary industry employment share are consistently negative across all periods, indicating an association between Shinkansen construction and secondary industry contraction. This phenomenon is consistent with Group A regions' high economic dependence on secondary industries and Japan's broader national trend toward tertiary industry development in recent years. Conversely, ATT estimates for tertiary industry employment share in Group A are consistently positive, suggesting that Shinkansen station construction may have supported the growth of services and other tertiary industries. This indicates that even in relatively remote areas, improving intercity accessibility can stimulate local service economy development, though the negative impact on the region's existing pillar industries cannot be overlooked.

In contrast, Group B, primarily comprising northern Kyushu and northern Chubu regions, exhibits higher urbanization rates, closer proximity to major metropolitan areas, and relatively developed secondary and tertiary industries. The opening of Shinkansen stations in these areas shows minimal or near-zero positive impact on secondary industry employment share, while tertiary industry ATT estimates are negative. This may reflect stronger competition or substitution effects in tertiary industries such as services in Group B regions with better transportation conditions and higher urbanization rates, where new stations did not generally generate net tertiary industry employment growth.

ATT estimates for employment rates in both regional groups are generally negative, indicating that overall employment rate growth in areas with Shinkansen stations did not significantly exceed that of matched control groups, and was even slightly lower than control groups. Possible explanations include labor outflow due to improved transportation or employment contraction in certain sectors during economic structural adjustment processes induced by Shinkansen construction. Meanwhile, unemployment rate impacts exhibit regional differences: Group A shows trends of increasing unemployment rates in some periods, possibly reflecting increased pressure from industrial structural adjustment or more severe labor outflow due to high-speed rail opening in these primary and secondary industry-dominated regions. Group B's unemployment rate ATT estimates are negative, indicating that despite the absence of net tertiary industry employment growth in Group B, the observed unemployment rate decline may reflect improved labor matching or absorption of unemployed individuals by other industries or neighboring metropolitan areas, benefiting from enhanced connectivity.

These findings emphasize the heterogeneous nature of high-speed rail infrastructure construction's employment impacts, suggesting that effects depend on local industrial structure, population dynamics, and economic context. Shinkansen station construction may promote service industry development in some regions while simultaneously creating pressure on manufacturing bases, leading to regional development imbalances. Therefore, when planning and implementing large-scale transportation infrastructure projects, policymakers should fully consider regional industrial structures and population characteristics, developing targeted regional development and labor market policies.

Additionally, compared to the literature referenced in this study, Jia et al. (2017) employed similar methodologies to analyze the impact of high-speed rail construction on regional economic growth in China. In contrast to our findings, while there may exist considerable variation in the magnitude of economic stimulation effects among different high-speed rail lines in China, with occasional instances of lines demonstrating negative impacts, the overall development of China's high-speed rail network has nevertheless generated positive promotional effects on the economic development of cities situated along these routes. This potentially indicates that high-speed rail plays distinctly different roles in economies of varying scales and at different developmental stages.

Furthermore, regarding the study's limitations, potential spatial spillover effects from long-distance labor mobility or broader inter-regional economic linkages cannot be completely ruled out. For example, employed populations in Shinkansen corridor regions may flow to more distant metropolitan areas, thereby affecting local employment structures. Due to data and methodological constraints, this study did not explicitly incorporate spatial lag terms or spatial weight matrices in the model to control for such spillover effects. Future research could employ spatial panel models (Spatial-DID) or spatial regression based on weight matrices to further explore the impacts of these long-distance spatial associations. In addition, we will perform industry-specific analyses while considering additional influencing factors, focusing on the specific impacts of high-speed rail development on industries such as manufacturing or services in surrounding areas. We will also include major cities in the scope of the investigation, separately calculating the effects of high-speed rail construction on metropolitan areas such as Tokyo as well as large regional cities such as Sendai.

CRediT authorship contribution statement

Jikang Fan: Writing – original draft, Formal analysis, Data curation. **Shintaro Terabe:** Writing – review & editing. **Hideki Yaginuma:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

This manuscript is original and has not been published or submitted elsewhere in part or in whole. And all authors have disclosed any potential conflicts of interest, and none were identified.

The corresponding author had full access to all the data in the study and had final responsibility for the decision to submit for publication.

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